

Scenario 18

A corporate network uses SDN. A new employee connects their laptop to the network for the first time. The network has a policy that all new devices must first be redirected to a registration portal.

Question: Describe the reactive flow installation process that enforces this policy.

Answer:

The reactive flow installation process redirects unauthenticated devices to a captive registration portal through the following step-by-step sequence:

1. Initial Connection & Packet Arrival:

The new employee's laptop connects to an SDN switch port and sends an initial network packet (e.g., an HTTP/DNS request).

2. Flow Table Lookup Miss:

The SDN switch inspects its hardware flow table. Because the laptop's MAC/IP address is unknown and unregistered, no matching flow rule exists in the switch table (a "table miss").

3. Packet-In Request to Controller:

Due to the table miss, the switch encapsulates the packet (or its header) into an OpenFlow `Packet-In` message and forwards it northbound over the control channel to the centralized SDN controller.

4. Policy Lookup & Reaction:

The SDN controller evaluates the packet against network policies and its device database. It identifies the device as unregistered and determines that the traffic must be redirected to the corporate captive portal.

5. Reactive Flow Mod Installation:

The controller sends an OpenFlow `Flow-Mod` message back down to the switch. This message installs a high-priority, temporary flow rule that matches the new laptop's ingress MAC/IP address and web traffic (TCP port 80/443). The rule's action rewrites the destination IP/MAC address to point to the registration portal (or redirects HTTP traffic to a captive web proxy).

6. Forwarding & Portal Redirection:

The controller issues a `Packet-Out` message for the initial buffered packet, and the switch forwards subsequent traffic according to the newly installed reactive flow rule. Any web request initiated by the employee automatically loads the registration portal.